

Statistical Pairs Trading System

A Python-based quantitative finance project focused on statistical arbitrage and pairs trading strategies using historical market data analysis.

Project Overview

This project explores correlated financial instruments and applies statistical methodologies to identify temporary market inefficiencies, spread divergence, and mean reversion opportunities.

Core Features

- Historical market data acquisition
- Correlation analysis between securities
- Spread and rolling statistical evaluation
- Z-score based signal generation
- Long/short trade simulation framework
- Data visualization and analytical charting
- Exploratory backtesting functionality

Technology Stack

Python, Pandas, NumPy, Matplotlib, yFinance, and quantitative analysis methodologies.

Professional Perspective

The implementation emphasizes analytical structure, statistical reasoning, and research-oriented experimentation commonly associated with quantitative finance workflows.

Disclaimer

This repository is intended strictly for educational, academic, and research purposes only and should not be interpreted as financial advice.

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